

Proposal Information Form

NOI	15720
PI	MD ARMANUZZAMAN
RA	

General Information:	
Official Project Title	SaTC 2.0: RES: VEST: Verifiable Trust Establishment onE
Official Project Dates	08/01/2027 to 07/31/2030
Is this a research project?	Yes ☒ No ☐
If yes, please select one type of research	Applied v
If no, please select one type of project	N/A v
If you selected "Instruction" or "Other" as the project type, please indicate whether your project includes any language or components related to diversity, equity, inclusion, or accessibility (DEIA)	
Is this a service award?	Yes ☐ No ☒
Is this a training award?	Yes ☐ No ☒

Abstract
In the box below, please provide a 50-300 word abstract. This abstract will be used for internal purposes only.
Embedded and cyber-physical systems increasingly rely on trusted execution environments, remote attestation, and compartmentalization, yet these mechanisms are applied once and in isolation: none confirms that a trusted domain contains only the resources an application needs, verifies execution stayed authorized afterward, or recovers selectively once compromised. This project develops VEST, a continuous trust lifecycle for resource-constrained embedded systems spanning three thrusts: application-specific trusted execution domains whose composition a remote verifier can confirm; scalable runtime attestation capturing control-flow, task, and interrupt behavior without evidence growing with execution duration; and interrupt-safe compartmentalization with authenticated data-flow policies that localize violations and selectively re-attest unaffected components. Together, these thrusts establish, verify, and preserve trust throughout execution rather than at a single point in time.

Personnel
In the box below, list all Co-PIs and Senior Personnel involved in the project, including both internal to UTEP and external to UTEP, regardless of their inclusion in the project budget. Provide the full name for each individual. For UTEP personnel, also include their UTEP email address.

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UTEP Personnel		
Using the table below, distribute 100% total participation across the PI, Co-PI(s), and any Senior/Key Personnel who are affiliated with UTEP. Do not include non-UTEP personnel. The participation percentages entered must sum to 100%. If more lines are needed, contact your RA for assistance.		
Role	Name	Participation %
PI	MD Armanuzzaman	100
Co-PI		0
Co-PI		0
Co-PI		0
Co-PI		0
Senior/Key Personnel		0
Senior/Key Personnel		0
Senior/Key Personnel		0
Senior/Key Personnel		0
Senior/Key Personnel		0
Total		0

Assurances	
Do you anticipate any contractual terms, conditions, and restrictions? Types of restrictions include but are not limited to: ITAR, EAR, OFAC and certain classifications	Yes ☐ No ☒
Is this a multidisciplinary proposal?	Yes ☐ No ☒
Does the proposal involve international collaborations, subawards, travel, student/ faculty exchange?	Yes ☐ No ☒
Does the proposal require additional space? Renovation/upgrades of current space? Do current facilities (infrastructure, equipment, power supplies, security doors/locks/ storage) support project activities?	Yes ☐ No ☒
Could this research lead to patentable invention?	Yes ☒ No ☐
Radioactive materials or radiation devices?	Yes ☐ No ☒
Recombinant DNA?	Yes ☐ No ☒
Human as research subjects?	Yes ☐ No ☒
Animals as research subjects?	Yes ☐ No ☒
Does your proposal include the use of human tissue and/or cell lines derived from human tissue (to include stem cells), a genome sequencer or software, or storage of genome sequencing data?	Yes ☐ No ☒
1. Does your proposal include biological agents?	Yes ☐ No ☒
a. If 1. is yes, will your proposal include any risk group 3 (RG3) or higher agents or any agents on the Federal Select Agents and Toxins list, including at amounts below the permissible toxin amounts?	Yes ☐ No ☒

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b. If 1. is yes, will your proposal include a risk group 2 or higher agent that involves manipulations that would be anticipated to result in any of the following experimental outcomes or actions: i. Enhance the transmissibility of the pathogen ii. Enhance the virulence of the pathogen iii. Enhance the immune evasion of the pathogen, such as by modifying the disrupt the effectiveness of pre-existing immunity via immunization or natural infection iv. Generate, use, reconstitute, or transfer an eradicated or extinct pathogen with pandemic potential	Yes ☐ No ☒
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Project Attributes: List all project locations, including any subcontractors, consultants, Co-PIs, and other collaborators.

Location Type	Usage	Location	Sub Location
On Campus			
On Campus			
On Campus			
On Campus			

Budget
Please enter a draft budget in the box below, including budget categories and estimated amounts. Examples include: PI summer salary (e.g., 2 summer months per year), undergraduate student support (e.g., two undergraduates per year), domestic travel (e.g., \$2,000 per year), and materials and supplies (e.g., \$2,000 per year).
PI: Per year 1 Summer month support for three years, total 3 months RA: 1 PhD RA support for three years RA: 1 Masters RA support for three years.